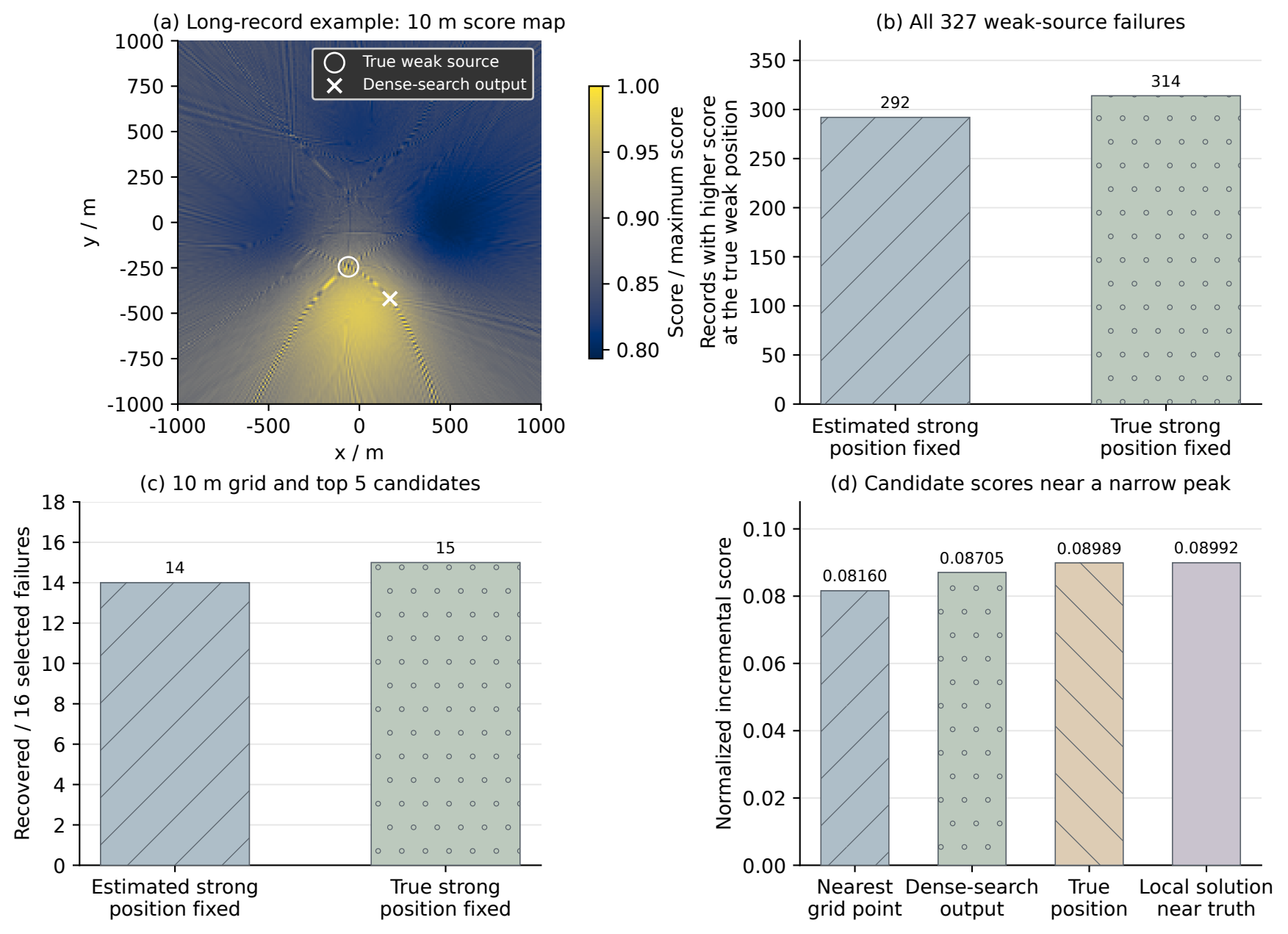

(a) Long-record example: 10 m score map
True weak source
Dense-search output
y / m
x / m
1000
750
500
250
0
-250
-500
-750
-1000
-1000
-500
0
500
1000
Score / maximum score
1.00
0.95
0.90
0.85
0.80
(b) All 327 weak-source failures
Records with higher score at the true weak position
350
300
250
200
150
100
50
0
292
314
Estimated strong position fixed
True strong position fixed
(c) 10 m grid and top 5 candidates
Recovered / 16 selected failures
18
16
14
12
10
8
6
4
2
0
14
15
Estimated strong position fixed
True strong position fixed
(d) Candidate scores near a narrow peak
Normalized incremental score
0.10
0.08
0.06
0.04
0.02
0.00
0.08160
0.08705
0.08989
0.08992
Nearest grid point
Dense-search output
True position
Local solution near truth